Towards Enhancing Driver's Perceived Safety in Autonomous Driving: A Shield-Based Approach

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Abstract. This paper proposes Personalized Perceived Safety Shielding (PPSS), designed to enhance perceived safety in reinforcement learning applications for autonomous driving. Utilizing a shield synthesis strategy, the framework ensures alignment with the driver's perceived safety preferences, derived from human driving data. Our framework intricately considers drivers' safety preferences, using real-world data for a personalized shielding strategy. This personalizing aligns with the human element in driving, addressing diverse safety perceptions effectively. The paper evaluates the proposed PPSS in terms of reflecting personal safety preferences and demonstrates its impact on the objective safety of driving. Through a detailed analysis of obstacle distances, the trajectories associated with the shield distance are examined. Preliminary results indicate that the trajectories generated by our framework are not only compatible with drivers' perceived safety needs but also maintain and enhance objective safety measures.

1 Introduction

With unprecedented technological advancements, reinforcement learning (RL) has been identified as an instrumental asset in addressing intricate practical issues, with its efficacy being profoundly noted in autonomous driving. These vehicles, capable of autonomously determining the optimal navigational actions amidst complex traffic, epitomize a revolution in enhancing transport safety and efficiency.

Yet, the real-world application of RL is not without challenges. There is a predominant focus on the learning algorithm's convergence, optimality, and data efficiency, often sidelining the agent's safety and correctness during learning and operation [1]. In the context of autonomous driving, ensuring that agents are equipped to avert dire outcomes like accidents is paramount. This has spurred

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a swift evolution in the field of safe RL, with a focus on methodologies ensuring the holistic safety of RL systems.

Alshiekh et al. [1] highlighted the efficacy of the shielding approach. Defined as a reactive system, a shield constantly monitors both the RL agent and its environment. It is tasked with the critical role of evaluating and, if necessary, substituting the agent's proposed actions with safer alternatives at each operational step [2]. However, a gap exists, especially where human trust and perceived safety are concerned. The deviation of shield-induced agent behaviors from human intuitive expectations exacerbates this gap. In this light, Xu et al. [3] coined the term "perceived safety," describing a state where both driver and passengers are imbued with a sense of security and comfort during the driving experience. It is an important issue to consider human safety preferences and intuition, as it will lead to improved practicality and acceptability of RL systems.

Therefore, RL that satisfies human perceived safety is also an important research direction. The work of D. Marta et al. [4] proposes a method of adjusting the shield synthesis strategy using human feedback to the output behavior of the agent to guarantee perceived safety. However, this approach requires iterative feedback and can be affected by subjective and unreliable human feedback. Also, Literature [5] proposes Human Intervention RL, which is a method that leverages human knowledge and feedback to support agent learning, has the challenge of the uncertainty regarding the number of queries to human experts occurring during the training process.

Against this background, we introduce a framework that employs actual human driving data to ascertain the optimal shield distance, tailored to individual perceived safety inclinations. By enhancing the shield synthesis strategy, our approach ensures a bespoke adaptation to drivers' safety perception, underpinned by in-depth analysis of driving trajectory data. The framework's efficacy was substantiated through rigorous testing, where individualized data derived from simulator-based driving scenarios was juxtaposed against RL. The advantages of our framework are (i) amplifying perceived safety by honoring individual distinctiveness and (ii) eclipsing the reliability and objectivity barriers associated with direct human feedback.

This paper is structured as follows. Section 2 introduces necessary background knowledge including basic reinforcement learning and safety reinforcement with shielding. Section 3 discusses different safety reinforcement learning approaches. Section 4 presents our framework called Personalized Perceived Safety Shielding. In Sect. 5, we describe how we evaluated our method, summarize the experimental setup and results, and discuss the results. Section 6 offers our conclusions and prospects.

2 Background

2.1 Reinforcement Learning

Reinforcement learning is a branch of machine learning. In this study, to model the fundamental decision-making problem for agents in reinforcement learning,

we assume the Markov Decision Process (MDP) framework, where the MDP is a five-tuple (S, A, T, R, γ) [6]. S represents the set of possible states in the environment, while A represents the set of actions available to the agent. $T(s'|s, a)$ is the transition function, defining the probability of the environment transitioning from state s to state s' when the agent takes action a in state s. $R(s, a, s')$ is the reward function, specifying the numerical rewards received by the agent when it takes action a in state s and transitions to state s'. A discount factor $\gamma \in [0,1]$ is a parameter that expresses how much the future compensation is discounted. The stochastic policy π of the MDP is the probability distribution of action probabilities for each state that defines the agent's behavior. The objective in setting the MDP is to find the optimal policy $\pi*$, where the agent chooses actions in each state according to π. And maximizes the expected prospective discounted reward [7].

2.2 Shielding

Shielding is an approach applied to traditional reinforcement learning, which is pre-computed based on an abstraction of the system's safety specifications and the agent's environmental dynamics. The shield is defined as $shield = \{(s, a)|P(s, a, k) > \delta\}$, where the exact probability is computed for how likely it is that running action a in some state s will violate specification φ within the next k steps, and action a is considered unsafe if the probability is greater than a threshold δ [8]. Applying a shield to a given specification by a correctly constructed reactive synthesis algorithm is called shield synthesis. The role of the shield is to prevent the agent from making incorrect decisions.

The shield meets two key requirements: ensuring correctness assurance and minimizing interference [9]. When it safeguards sequential decisions, it guarantees safety throughout the learning and execution phases, providing continuous safety as long as it's active. It's designed to make minimal adjustments to the agent's decision-making, intervening only when safety is at risk.

One of the remarkable strengths of the shield is its capacity to separate concerns and scale effectively [9]. They operate independently of the learning algorithm, maintaining accuracy regardless of the chosen learning method. This clear distinction in roles allows the learning agent and the shield to focus on their respective tasks. In terms of scalability, shields can assess safety within specific parts of the system, using a partial model rather than the entire system. This approach enhances safety assessment for specific system components while leveraging the overall model for improved performance.

3 Related Work

Various approaches have been proposed to address perceived safety concerns in RL. Behavior Cloning [10] is a supervised learning technique that learns combinations of states and actions from expert behaviors. However, while behavior cloning is an attractive and simple approach, it has the problem of covariate

shift and tends to require a lot of data to address this issue [11,12]. Also, inverse reinforcement learning [13] requires reinforcement learning in an inner loop and is very expensive to execute because it calculates the reward function and goals pursued by the agent in reverse from the agent's behavioral data. Literature [14] operates imitation learning, which is a machine learning technique that learns human behavior and decision-making. The learning policy simply mimics the data, making it difficult to understand the strategy behind it, which can lead to unsafe behavior. Literature [5] proposes Human Intervention RL (HIRL), which is a technique that incorporates human knowledge and feedback to help agents learn, but it is unclear how many queries to human experts occur during the training process. Literature [15] proposes Convergent Actor-Critic by Humans (COACH), an algorithm that learns from policy-dependent feedback and converges to a local optimum but requires frequent queries to humans during training, which may limit its use in real-world situations. Literature [6] proposes an approach that divides large tasks into smaller subtasks and incorporates basic behavioral primitives to solve each subtask but requires humans to specifically select primitives, which may be a limitation on flexibility. Additionally, literature [4] proposes enhancing the strategy of shield synthesis through continuous human feedback on the actions of the agent to guarantee safety. However, these methods pose challenges due to their reliance on time-intensive and unreliable human subjective feedback.

Addressing these challenges, this paper proposes a framework that satisfies different perceived safety among individuals by determining the appropriate distance from actual human driving data and applying this value as a parameter of the shield (defined as the shield distance). In other words, this framework enables shield synthesis to be personalized to the driver's safety perception needs through the analysis of driving trajectory data.

4 Personalized Perceived Safety Shielding

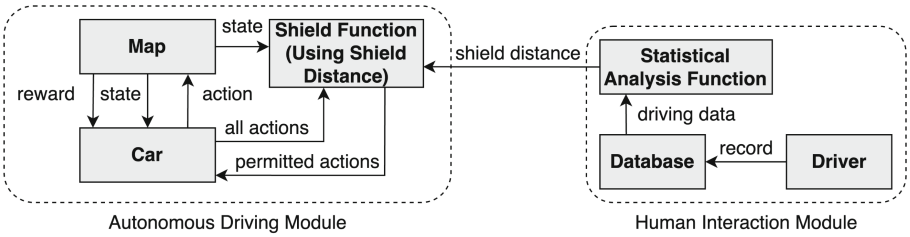

Fig. 1. Framework overview

We propose a framework called Personalized Perceived Safety Shielding (PPSS), which incorporates shielding based on historical human driving data into

reinforcement learning and aims to satisfy human perceived safety by treating it as a numerical value. As represented in Fig. 1, PPSS consists of the autonomous driving module that reflects drivers' safety preferences using the shield function and the shield distance when the agent drives, and the human interaction module that uses the statistical analysis function of driving data collected from driver to personalize the minimum distance required to satisfy the driver's perceived safety as the shield distance and transfers it to the autonomous driving module.

4.1 Autonomous Driving Module

The role of the autonomous driving module is to realize reinforcement learning using the shield function. The dynamics of reinforcement learning are modeled by a Markov Decision Process. The agent's primary goal is to find an optimal policy that chooses the best action a for a given state s, and this optimal policy maximizes the agent's expected cumulative reward. Here, S represents the set of possible states in the environment and A refers to the set of actions that the agent can choose.

The shield function $\Psi : s \times \mathbb{R} \rightarrow \{a|P(s,a) > \delta\}$ takes the agent's states and shield distance θ and maps them to a set of permitted actions ($\mathbb{R}$ is a set of real number and δ is the threshold that satisfies safety).

The purpose of this function is to restrict the range of applicable actions in each state and serves to provide a set of actions that are considered permissible for the agent. The agent selects the optimal action from the set of permitted actions. $A_{pmt}(s,\theta) := \Psi(s,\theta) \subseteq A$ received from the shield function and applies it to the environment; A_{pmt} guarantees that the agent satisfies the safety distance and the shield distance θ is the value received from the human interaction module.

4.2 Human Interaction Module

The human interaction module determines the shield distance reflecting the perceived safety. Specifically, the shield distance θ is a number calculated from human driving data and is defined as the minimum distance a moving vehicle should be away from surrounding obstacles to be perceived as safe by humans. This shield distance is used in the shield function of the autonomous driving module.

The database collects and stores the driver's driving trajectory data. The statistical analysis function first extracts the distance between the vehicle and the nearest obstacle at each time step from the data stored in the database. It then calculates the appropriate shield distance based on that information and outputs it to the shield function. The statistical analysis function can be defined using Eq. (1).

$$\theta = \text{percentile}(L_d, 25) \tag{1}$$

where θ stands for shield distance, L_d represents the set of distance information in the driver's trajectory. We explain why we used the value of 25 in Eq. (1).

The assumption is that people are conscious of driving in the middle of the road, which is the safest place to drive. The driving data obtained in this experiment is characterized by the fact that most of the data are distributed at a distance of 4 m (80% of the data range) or more from obstacles, and the closer the distance, the smaller the number of data. From this relationship between distance and number of data, it can be inferred that humans feel "safe" in most cases while driving, and "dangerous" in only a few cases. In terms of perceived safety, it can be considered that the percentage of data far from obstacles is high no matter who is driving, while data close to obstacles is more valuable in that it is easily influenced by the driver's driving and reflects human preferences for perceived safety. Therefore, this paper uses the first quartile, a measure of data variability in statistics, for the statistical analysis function. Assuming that the percentage of cases in which humans feel "unsafe" while driving is less than 25%, the 25th percentile value can be used to apply the shield distance θ as the minimum distance a driver should be away from obstacles to always satisfy perceived safety while driving. Also, by using the first quartile value, the value can be determined using an even distribution of data independent of road and driver.

5 Evaluation

The research questions of the evaluation are as follows:

RQ 1 To what extent does automatic driving with an applied PPSS align with individual drivers' safety preferences?

RQ 2 How does the incorporation of a shield, designed to meet drivers' perceived safety, impact the objective safety of automatic driving?

This research aims to ensure that automated driving with PPSS meets the driver's perceived safety while aligning with their driving. In Experiment 1, we evaluate this consistency by examining obstacle distances from driving data before and after applying our framework. In addition to addressing RQ1, we also consider objective safety. Experiment 2 assesses the impact on objective safety by comparing the number of steps to collision. A higher number of steps post-framework implementation indicates greater objective safety, providing more reaction time before a potential collision.

5.1 Experiment Setting

The flow of experiment 1 is to collect the driver's driving data, determine the shield distance, acquire automated driving data before and after the application of PPSS, calculate the distance to obstacles at each step, and evaluate the similarity of the histograms of the distance data.

A 2D map for the RL environment and a 3D map for collecting driving data using a driving simulator are prepared as shown in Fig. 2. The two maps have the same shape: a mouth-shaped road with a road width of 10 m, bounded by a

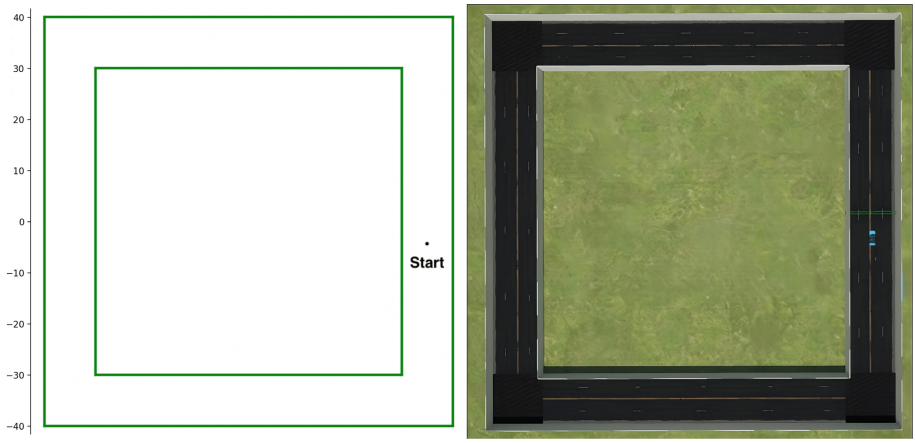

Fig. 2. 2D map (in RL) and 3D map (in UE4)

square wall with an inner road frame of 60 m, and a square wall with an outer road frame of 80 m.

Reinforcement learning uses the Deep Deterministic Policy Gradients (DDPG) algorithm. The input data for the agent is sensor data and velocity information in 15 directions, and the output is angle and acceleration information. The experiment was conducted for 50 episodes, with an upper limit of 1500 steps in each episode. Agents were pre-trained, and the agent's trajectory was recorded with the condition of applying or not applying PPSS at run time.

In the driving simulator environment, a map was created using UnrealEngine4 (UE4), and a human controller was used to collect driving trajectories. The controller allows a human to control the vehicle, especially acceleration, braking, and left/right handling, which is nearly identical to how the vehicle behaves in real driving. Participants collected 50 episodes of data, with one episode being about a car colliding with a wall. We enrolled 11 participants (9 males, and 2 females) from our universities. They had normal hearing and either normal vision or corrected vision. Their ages ranged from 22 to 25, with an average age of 23.7 (SD 0.9). Participation was voluntary with no financial incentives. As for information about the participants' driving, all of them have obtained a driver's license. Since all participants are students, the elapsed time since obtaining their driver's license is short, ranging from a minimum of 2 years to a maximum of 6 years, with an average of 4.18 years. In terms of driving frequency, three participants have not driven at all since obtaining their driver's license, three have driven a few times every few months, and the remaining five drive at least a few times a month. Figure 3 shows the view the experimenter sees while driving and experimenting with the controller.

The trajectory data was converted into the histogram data set representing the distance to the nearest obstacle for comparison. The objects to be

Fig. 3. Scence of the user experiment

compared are histograms of distance data to obstacles for human actual driving, and automated driving before and after applying our framework.

We evaluate histogram similarity using three methods: overlap similarity, cosine similarity, and the chi-square statistic. Overlap similarity measures common bin frequencies, with larger values indicating greater similarity. Cosine similarity treats the histogram as a vector and considers shape similarity, with higher values indicating greater similarity. The chi-square statistic assesses frequency differences between corresponding bins, with smaller values indicating greater similarity. We use all three of these methods because each provides distinct insights from different perspectives, offering a comprehensive assessment of histogram similarity. If the histograms after applying PPSS demonstrate higher similarity than those before, it underscores the effectiveness of our framework in meeting drivers' perceived safety needs. When histograms A and B are considered as separate data sets and the values of each bin i are represented as a_i and b_i respectively, the three similarity measures are defined as follows:

$$\text{Overlap Similarity}(A, B) = \sum_i \min(a_i, b_i) \tag{2}$$

$$\text{Cosine Similarity}(A, B) = \frac{\sum_i a_i b_i}{\sqrt{\sum_i a_i^2}\sqrt{\sum_i b_i^2}} \tag{3}$$

$$\text{Chi-Square Distance}(A, B) = \sum_i \frac{(a_i - b_i)^2}{a_i + b_i} \tag{4}$$

The flow of experiment 2 is to evaluate objective safety by calculating and comparing the number of steps in the automated driving data before and after applying PPSS obtained in experiment 1. We compare the maximum, minimum, and average number of steps until the end of 50 episodes (collision or achieving 1500 steps set as the maximum number of steps) concerning the driving data of

the agent after applying PPSS with shield distance for each of the 11 participants and before applying it.

5.2 Experiment Result

The results of experiment 1 are presented in Table 1. It shows the shield distance calculated from the driving data of 11 participants and compares the three types of similarity before and after the application of PPSS using those values. The gray background indicates a high degree of similarity. The overlap similarity and chi-square statistic showed a higher degree of similarity after application for all participants, while the cosine similarity showed a higher degree of similarity after application for 9 participants. Specifically, the improvement rate for overlap similarity, using 11 data sets, averaged 24.4%, with a minimum of 1.23% and a maximum of 43.0%. The chi-square statistic also shows an average improvement of 56.0%, with a minimum improvement of 12.4% and a maximum improvement of 78.7%. As for the cosine similarity, the data of the nine individuals whose similarity was higher after the application showed an average improvement rate of 40.0%, a minimum of 2.23%, and a maximum of 84.8%.

Table 1. Value of shield distance and comparison of histogram similarity

	1	2	3	4	5	6	7	8	9	10	11
Shield Distance[m]	2.12	2.59	2.32	2.26	1.96	2	2.22	2.32	2.32	2.18	2.24
Overlap(Before)	0.734	0.816	0.806	0.638	0.584	0.401	0.513	0.582	0.668	0.543	0.613
Overlap(After)	0.837	0.826	0.845	0.798	0.765	0.508	0.683	0.832	0.849	0.723	0.79
Cosine(Before)	0.898	0.937	0.939	0.735	0.655	0.263	0.466	0.571	0.781	0.546	0.702
Cosine(After)	0.918	0.916	0.922	0.902	0.879	0.486	0.757	0.899	0.941	0.812	0.89
Chi-square(Before)	0.185	0.097	0.105	0.346	0.424	0.883	0.607	0.465	0.295	0.527	0.384
Chi-square(After)	0.081	0.085	0.075	0.107	0.135	0.592	0.248	0.099	0.067	0.189	0.117

The results of experiment 2 are shown in Table 2 and Fig. 4. The maximum, minimum, and average values of the number of steps in the automated driving data using the shield distance determined from the 11 participants were obtained, and the average of each is shown as the result of the number of steps in the automated driving data after the application of PPSS in Table 2. The gray background indicates a large number of steps. The number of steps for each participant's driving data after the application of PPSS and before the application is shown in Fig. 4, where the number on the horizontal axis indicates the user reflected in the determination of the shied distance, and "Before" indicates before PPSS was applied. The results show that in all three comparisons, the driving data after applying PPSS had more steps for all participants. Specifically, considering the number of steps after application, the maximum number of steps is 2.28 times, the minimum number of steps is 2.51 times, and the average number of steps is 2.24 times, showing a significant increase over the number of steps before application for all three evaluation targets.

Table 2. Comparison of the number of steps in automated driving data before and after applying PPSS

	Automated driving data Before applying PPSS	Automated driving data After applying PPSS
Maximum Number of Steps	626	1430
Minimum Number of Steps	350	878
Average Number of Steps	527	1180

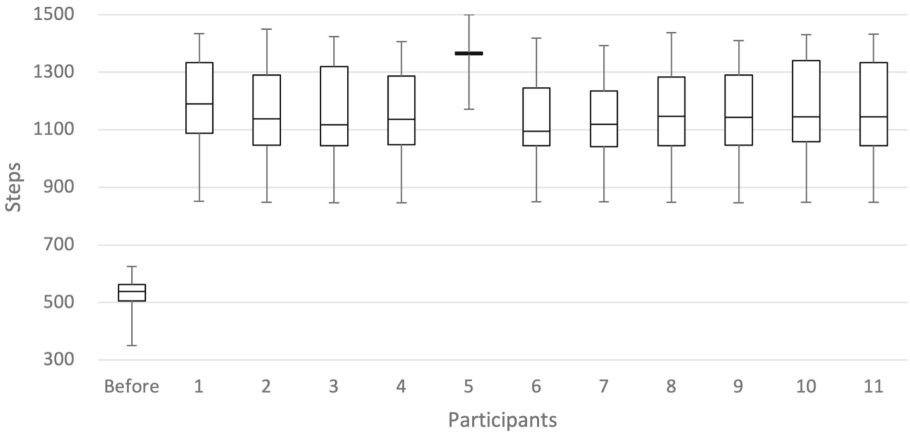

Fig. 4. Comparison of the number of steps for all participants

5.3 Discussion

For RQ1, it was clear that automated driving after applying our framework was closer to the actual driver driving and could better reflect the driver's perceived safety. The reason is that in Experiment 1, the similarity of driving data after applying PPSS was higher for most participants. Two participants had higher cosine similarity before the application of PPSS, both with a similarity of 0.9 or higher. In the future, it will be necessary to investigate in depth whether this is an effect of the statistical analysis function, the driver's driving itself, the shape of the map, or what it depends on. Regarding RQ2, our framework for satisfying perceived safety did not negatively affect objective safety, but rather confirmed that it makes driving safer.

There are three main points of discussion: how to determine the shield distance, the shape of the map, and the type of information obtained from the driving simulator.

First, regarding the method of determining the shield distance, in this paper, the statistical analysis function assumed the first quartile as the shield distance based on the driving data, and we believe this method was appropriate. The minimum distance to surrounding obstacles while driving a vehicle is necessarily

closer when driving on a curve than when driving on a straight road. On roads with a mixture of straight and curved roads, such as the real-world roads and the maps used in the experiment, the use of the first quartile in terms of extracting data on distances closer to obstacles that are more susceptible to human driving is effective in achieving the minimum distance that satisfies individual perceived safety.

Next, The shape of the map was used in this experiment with the same shape and size for both the 2D and 3D maps. This setup was very convenient for this experiment, and actual roads are rarely as clean and square as in the experiment. Therefore, for future research, it is necessary to prepare maps of different shapes and to verify whether the method is effective when applied to maps of different shapes. It is also necessary to confirm the validity of the method for determining the shield distance using the first quartile in maps of different shapes.

Finally, regarding the type of data to be obtained from the driving simulator, this time only position information was collected and used as distance data from surrounding obstacles. In the future, we believe that collecting not only position data but also additional data such as vehicle speed and vehicle angle will lead to results that better satisfy the driver's perceived safety.

6 Conclusion and Future Work

We presented the Personalized Perceived Safety Shielding in this paper, an innovative methodology that capitalizes on the analysis of historical human driving data to refine the shield distance, thereby offering an individualized enhancement of perceived safety. The initial assessments attest to a marked improvement in drivers' safety perception with the integration of our tailored shield. Nevertheless, we recognize the imperative of further rigorous validations to affirm the efficacy and reliability of this approach comprehensively.

Looking ahead, our future road map encompasses the augmentation of the shield's synthetic statistical analytical prowess, aiming for a model that transcends the constraints of road geometry. Additionally, we are committed to assimilating a broader spectrum of driving information, an initiative geared towards a more nuanced and comprehensive satisfaction of human-perceived safety requisites.

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